

Gaurav Singh

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Professional profile

Software developer and designer with expertise in **human-centred design and experience design**, and a background spanning engineering and design schools. Build payment systems in **C#/.NET and TypeScript/Vue**, alongside published applied-ML research. Previously headed the M.Des Design Computation programme at Srishti Manipal Institute and led development of an agricultural traceability demonstrator.

Engineering and design skills

Software engineering: C#/.NET, TypeScript, Vue, React, Node.js/Express, Python/Flask, REST APIs and SQL.

Design: human-centred design, experience design, interaction design, physical computing and human-computer interaction; guiding designers from brief interpretation through delivery.

Testing and delivery: automated testing, Docker, Linux, Git, GitHub Actions and Azure DevOps.

ML experimentation: PyTorch, PyTorch Lightning and scikit-learn; **Optuna for hyperparameter optimisation** and **Weights & Biases** for experiment tracking.

Experience

2025-present

Software Developer, Pay Advantage, Australia

- Develop payment-platform software covering recurring billing, direct debit, subscriptions, bank-account runs, card payments, hosted payment pages, and API integrations.
- Work with C#, .NET, TypeScript, Vue, Azure DevOps, and REST APIs.

Selected projects

AATP: agricultural traceability demonstrator

2024

Master's project, Griffith University · Supervised by the team at Anonymo Labs

- Led development and **built the complete demonstrator** using Python/Flask, VON Network and ACA-Py, integrating digital identity wallets, credential schemas and web and command-line interfaces.
- Linked production, packaging and transport credentials by product ID for traceability; debugged agent-ledger connectivity and Indy/OpenSSL dependencies across macOS, Ubuntu and Docker.

Machine learning for auxetic deformation

Published 2025

- Co-developed a workflow combining finite-element simulation data, K-means image labelling and logistic regression to automate deformation-mode detection in re-entrant honeycomb structures.
- Detected **four of six deformation modes** in the study, establishing the method's demonstrated scope. Published in *International Journal of Protective Structures*.

Heavy-transport decarbonisation modelling

Published 2026

- Co-developed a framework linking Australian emissions accounting and fleet-transition economics with nonlinear technology-cost learning and grid decarbonisation.
- Evaluated emissions, ownership costs and ROI across **270 scenarios over 25 years**. Sensitivity analysis identified diesel price, electricity price and utilisation as dominant economic drivers.

Earlier experience

- 2017-2022 **Faculty Member and Researcher, Srishti Manipal Institute**, Bangalore, India
- Headed the Master of Design in Design Computation.
 - Set up the Experimental Maths Lab.
 - Taught interaction design and human-computer interaction alongside programming, algorithms, machine learning and physical computing.
- 2018-2022 **Design Copilot (Part-time)**, Topcoder, Remote
- Ran design challenges for Fortune 500 clients and guided distributed groups of designers from brief interpretation through final delivery, after several years competing on the platform.

Education

- 2023-2025 **Master of Information Technology**
Griffith University, Gold Coast, Australia; GPA 6.31/7
Thesis: "Enhancing Climate Forecasting with Advanced Generative Models for Synthetic AQI Data", supervised by Dr. Elizabeth Chang.
- 2013-2015 **Advanced Diploma, Innovation & Experience Design**
Srishti Institute of Art, Design and Technology, Bangalore, India
- 2008-2012 **B.Tech, Computer Science & Engineering**
JNTU Anantapur, India; First Class with Distinction

Selected publications

- A Framework for Visual-Mathematical Literacy in Applied Machine Learning: Why Representation Choices Shape What Models Can Learn. G. Singh, R. S. Dhari. Machine Learning: Engineering, 2026. <https://doi.org/10.1088/3049-4761/ae7df3>
- Techno-Economic Pathways Modeling and Nonlinear Optimized SEEA-ROI Longitudinal Dynamic Simulation for Decarbonizing Australian Heavy Transportation Systems. G. Singh, E. Chang, Y. Karaca. Fractals, 2026. <https://doi.org/10.1142/S0218348X26400633>
- Automated detection of deformation mechanisms in re-entrant honeycomb auxetics using machine learning. G. Singh, R. S. Dhari, Z. Javanbakht. International Journal of Protective Structures, 2025. <https://doi.org/10.1177/20414196241281069>
- ReRide: A Bike Area Network for Embodied Self-monitoring during Motorbike Commute. N. Bagalkot, G. Singh, V. Rath, T. Sokoler, A. Shukla. TEI 2019, 443-450, 2019. <https://doi.org/10.1145/3294109.3300986>

Selected recognition

- 2024 **Griffith Award for Academic Excellence** (Griffith University)
- 2024 **STEM Professional Gold Achievement** (Griffith University)
- 2017 **Topcoder Open Design Champion** (India Regionals)

Portfolio: <https://gaurav-singh.info>

Blog: <https://barelydiscernible.bearblog.dev/>